Prompt 1 : For research

You are an expert AI systems architect and hackathon strategist. Analyze and conduct deep-dive technical research for the two hackathon tracks provided below, extracting the core technical constraints, judging priorities, and state-of-the-art implementations.

Context
1. Track A: Rime Hackathon Challenge ("Build a Voice-Native Product")
- Objective: Build a working voice-first prototype where speech is strictly essential (not a chatbot with a play button).
- Core requirement: Solve one hard voice engineering failure mode (e.g., perceived response time, full-duplex interruption & recovery, conversation continuity during tool calls, pronunciation/prosody, telephony/packet loss).
- Tech stack requirements: Rime TTS as primary spoken output, LiveKit Agents (recommended transport/orchestration), reproducible acceptance tests in `RIME_EVIDENCE.md`.
- Judging: Problem & necessity of voice (25%), Hard voice engineering (25%), Rime integration (20%), Evidence & reproducibility (20%), Demo clarity (10%).

2. Track B: Pathway Track ("Explain the Frontier" - NeurIPS 2026 Education Track format)
- Objective: Build an interactive, self-contained educational artifact explaining a frontier AI concept (Linear Attention, Synaptic Plasticity, Test-Time Adaptation, Latent Reasoning, Post-Transformers) with a mandatory, technically rigorous module connecting it to Pathway's Dragon Hatchling (BDH) or BDH-CQ architecture.
- Core requirement: One falsifiable central claim; interactive substrate (real live computation/toy model, not scripted animation); truth-beside-estimate feedback under 1 second; 500–950 word one-page concept briefing PDF.
- Judging: Technical correctness & depth (25%), Ownership & live defense (15%), Learning effectiveness (15%), Interactive substrate (15%), BDH integration (10%), Craft/provenance (10%), One-page summary (10%).

Tasks
1. Track A Research:
  - Identify 3 distinct, high-impact user personas and workflows where removing voice breaks the product (e.g., hands-busy surgical/field repair logging, dispatcher telephony, real-time debate/coaching).
  - Detail the end-to-end technical pipeline for LiveKit Agents + Whisper/Deepgram STT + LLM streaming + Rime TTS WebSocket streaming.
  - Explain how to formally engineer and measure an acceptance test for: (a) sub-400ms time-to-first-audio (TTFA) and (b) audio cancellation/state rollback during interruption.
2. Track B Research:
  - Break down the mathematical and architectural fundamentals of Pathway's BDH (attention as Hebbian synaptic updates, sparse non-negative activations) vs. BDH-CQ (latent recurrent reasoning without CoT).

- Review 3 candidate frontier concepts from the approved list (e.g., "Linear Attention vs. Softmax Attention", "Test-Time Adaptation: HRM/TRM vs. BDH-CQ", "Latent-Space Reasoning vs. Verbalized CoT") and identify which concept has the sharpest interactive potential.
- Outline the best interactive web architecture (e.g., Pyodide/ONNX in WebAssembly, React + Three.js/Canvas, observable state) to ensure immediate (<1s) client-side feedback without costly backend GPUs.

Prompt 2: For selecting better PS.

You are a competitive hackathon judge and senior technical lead. Evaluate Track A (Rime Voice-Native Product) and Track B (Pathway Explorable Explainer on BDH) to recommend the optimal choice for a hackathon team, complete with a risk assessment and difficulty breakdown.

Inputs to Weigh
- Team Profile & Strengths: Full-stack development, ML/math depth, frontend interactivity/visualization skills, audio streaming/WebRTC familiarity.
- Submission Constraints:
  - Rime: 4–5 min recorded demo video, working repository, `RIME_EVIDENCE.md` test script, LiveKit/Rime secret preflight check, production API integration.
  - Pathway: Public web demo (zero-login), interactive substrate with live toy model, 500–950 word concept summary PDF, citations of 3+ papers (2022–2026), live defense readiness against technical edge cases.

Deliverables
1. Side-by-Side Scoring Breakdown:
   - Compare both tracks across: Implementation velocity, Debugging difficulty, Risk of demo failure during judging, Tooling maturity, and Evaluation subjectivity.
2. Failure Modes & Blockers:
   - For Track A: Detail the hardest engineering hurdles (e.g., audio pipeline drift, WebSocket race conditions on user barge-in, latency budget over telephony vs. web, edge-case pronunciation failures).
   - For Track B: Detail the pedagogical and technical pitfalls (e.g., accidentally building a "scripted toy" instead of a reactive substrate, overcomplicating math so the learner gets lost, weak or bolted-on BDH linkage, math/algorithmic inaccuracies penalizing 25% of points).
3. Decision Decision Tree:
   - Provide concrete criteria: "Choose Track A if your team has [X skills/assets]; choose Track B if your team has [Y skills/assets]."
4. Final Recommendation:
   - Give a direct, justified recommendation on which track has the highest ceiling to win an exceptional score within a limited hackathon sprint.

PROMPT 3: FOR SOLUTION ASSEMBLY

You are a principal software engineer and AI researcher. Generate complete, production-ready solution blueprints for BOTH Track A (Rime Voice-Native) and Track B (Pathway Frontier Explainer). Each solution must be concrete, architecturally sound, and directly aligned with the judging rubrics.

Approach 1: Track A (Rime Voice-Native Product)
1. Product Concept & "Necessity of Voice":
   - Name a razor-focused use case where a screen or text interface fails (e.g., hands-busy industrial safety inspector or live multilingual dispatch).
2. Selected Voice Engineering Challenge:
   - Focus on Full-Duplex Interruption & Conversation Continuity during Long-Running Tool Calls.
3. System Architecture & Tech Stack:
   - Define the exact pipeline: LiveKit Agents framework, STT endpoint, streaming LLM orchestrator with function-calling fences, and Rime TTS WebSocket streaming.
   - Diagram the state machine handling: User speaking -> VAD trigger -> prompt LLM -> initiate slow tool call -> user barge-in/cancellation -> purge queued Rime audio -> state rollback.
4. Acceptance Test Specification (`RIME_EVIDENCE.md`):
   - Step-by-step test script injecting simulated latency into a background tool call while the agent speaks, asserting audio playback stops within <200ms and stale payloads are discarded.
5. 4-Minute Demo Script:
   - Turn-by-turn script covering: problem pitch (30s), normal flow (90s), deliberate stress/interruption test (90s), metric/evidence display (30s).

Approach 2: Track B (Pathway Interactive Frontier Explainer)
1. Concept & Falsifiable Claim:
   - Concept: "Recurrent Latent Reasoning vs. Chain-of-Thought" or "Linear Attention as Synaptic Memory".
   - State the single falsifiable claim (e.g., "Latent recurrent refinement increases puzzle-solving accuracy under fixed token budgets without emitting intermediate reasoning tokens, but degrades under unbounded constraint interference").
2. Interactive Substrate Design:
   - Specify a lightweight, runnable toy system running in the browser (e.g., a tiny 2-layer BDH-CQ or linear attention network in WebAssembly/TypeScript).
   - Define exact concept sliders/controls (e.g., recurrence depth K, context length, weight decay λ) and instantaneous visualization (hidden state trajectory, output token probability distribution vs. ground truth).
3. The BDH/BDH-CQ Module:
   - Outline the mathematical exposition connecting the toy model to the published BDH/BDH-CQ papers (equations for synaptic weight updates $W_{t} = W_{t-1} + \eta x_t y_t^T$ and non-negative activations).
4. One-Page Concept Summary Blueprint:

- Provide a section-by-section breakdown for the 500–950 word briefing PDF: problem motivation, technical mechanism, comparative analysis table (Transformer vs. Mamba vs. BDH-CQ), evidence classification, and known architectural bounds.